

INDIAN STATISTICAL INSTITUTE

Project Linked Person — Selection 2026–2027

Subject: Systems Support (Software, Hardware and Web)

Part B — Practical Examination

Total: 14 + 16 + 10 = 40 marks Duration: 90 minutes

EXAMINATION INSTRUCTIONS

1. You have root on a virtual machine. It is connected to no real network. Log in with `ssh candidate@192.168.56.10`.
2. Each question has four parts — *the problem, how to reproduce it, what to fix, and how to check your work*. Run the checks yourself; you should not have to guess whether you are finished.
3. **The *Constraints* of every question are marked. Breaking a constraint loses marks even when the symptom goes away.** Examples: `chmod 777`, pinning names into `/etc/hosts`, reformatting a filesystem.
4. Keep `~/FIXLOG.md` up to date as you go: symptom, cause, what you changed, and how you verified it. It is marked, and there is a short oral discussion of your work afterwards.
5. You are not expected to know all of this in advance. Partial credit is real: a fault you diagnose correctly and explain, but cannot finish, is worth more than one you paper over.

NOTE: `man`, `--help`, `journalctl` and the logs under `/var/log` are all available on the machine. Your commands are recorded. Questions may be attempted in any order.

Q1. Disk full on the lab file server.

[14 marks]

Reported by Debarshi Roy, research scholar: “I cannot copy anything into `/srv/data`, it says the disk is full, and nothing I can see in there accounts for that much space. Submitting a job into `/srv/spool` *also* says the disk is full, but `df -h` shows that one has plenty of space left. That part makes no sense to me.”

Background: this server has three separate disks. `/srv/data` and `/srv/spool` look like ordinary folders, but each one is really its own small disk, attached at that path (a *mount point*). The rest of the system lives on a third, much larger disk.

<code>/</code>	the system disk	about 24 GB
<code>/srv/data</code>	the project data disk	1 GB
<code>/srv/spool</code>	the job-submission disk	48 MB

Each disk fills up on its own: one being full says nothing about the others. `df -h / /srv/data /srv/spool` prints one line per disk.

The two complaints look alike — both end in “No space left on device” — but they are **two unrelated faults on two different disks**, with different causes. Fixing one does nothing for the other. Tackle them in either order; each is marked on its own.

(a) /srv/data is full **[11 marks]**

/srv/data is at 100%, and the files you can see there do not add up to the space reported as used.

Reproduce:

```
df -h /srv/data          # 100% used
sudo du -sh /srv/data    # much smaller than df says is in use
```

Fix: bring /srv/data back to **under 10% used** (the Use% column of `df -h`). There is more than one reason this disk is full: freeing the largest obvious files will not get you there on its own. Treat any difference between what `df` says is used and what `du` can account for as something you still have to explain.

Check:

```
df -h /srv/data          # Use% under 10%
ls /srv/data/current     # five survey-block CSV files, still there
```

(b) /srv/spool refuses new files **[3 marks]**

New files cannot be created on /srv/spool, *although `df -h` reports free space on it*. Something other than free space has run out. This has nothing to do with /srv/data.

Reproduce:

```
df -h /srv/spool         # shows space free
touch /srv/spool/incoming/testjob # "No space left on device"
```

Fix: make /srv/spool able to accept new files again. It is *not* short of space: freeing bytes there will not help. A filesystem can run out of more than one kind of resource, and `df` has another mode that reports the other one. Then, in ~/FIXLOG.md, explain in two or three sentences why /srv/spool reported “No space left on device” while `df -h` showed free space.

Check:

```
touch /srv/spool/incoming/t && echo OK && rm /srv/spool/incoming/t
```

Constraints

- Nothing under /srv/data/current may be deleted or changed. It is live survey data with no second copy on this machine.
- Do not reformat, recreate, resize or unmount either disk.
- /srv/data/archive/README.txt says what is safe to remove there. Read it before deleting anything.
- Rebooting clears one of these faults. On a real file server at 11am you would not have that option; treat it as unavailable.

Q2. Shared project folder unusable. **[16 marks]**

Reported for the ACMU Lab group: “Buddhadev cannot save anything into **shared/**, and cannot open **notes.md** at all. When I create a file there, the others cannot edit it — it comes out belonging to my own refused. And Chandrima from the internal audit cell needs to *read* that folder for the annual review. He cannot see anything.”

Background: who is who. `/srv/projects/acmu/` (and `shared/` inside it) belongs to the `acmu` group. Anyone who is not a member or the auditor should have no access at all.

<code>arnab</code>	member of <code>acmu</code>	read and write everything
<code>buddhadev</code>	member of <code>acmu</code>	read and write everything
<code>chandrima</code>	auditor, not a member	read everything, write nothing

To see the current state:

```
cd /srv/projects/acmu
stat -c '%a %U:%G %n' . shared shared/notes.md
id buddhadev; getfacl shared; sudo cat /etc/sudoers.d/acmu
```

The complaints overlap, but underneath them are **five independent faults**, each with its own cause and fix. Fixing one does nothing for the others; each is marked on its own. Tackle them in any order. Test as the users named, not as root — root can do everything, so a check run as root proves nothing.

(a) Buddhadev is no longer in the `acmu` group **[4 marks]**

Reproduce: `id buddhadev` — `acmu` is missing. *Fix:* put him back, without changing any of his other groups.

```
id -nG buddhadev | grep -qw acmu && echo "buddhadev is a member"
```

(b) The `shared` folder does not let the group write **[7 marks]**

Members of `acmu` cannot create files in the folder at all. This is about the permission bits on the two directories, `/srv/projects/acmu` and `shared/`. Arnab is still a member, so test as her.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1 # Permission denied
stat -c '%a %U:%G %n' /srv/projects/acmu /srv/projects/acmu/shared
```

Fix: both directories must let the `acmu` group read, write and enter them, and give everyone else no write access. Remember that a directory needs its execute bit before anyone can go into it at all.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1 && echo "write OK"
sudo rm -f /srv/projects/acmu/shared/t1
```

(c) `notes.md` has become private **[3 marks]**

`shared/notes.md` is the group's working file, but now belongs to one person and only she can read it. *Fix:* group `acmu`, readable and writable by the group, no access for anyone else.

```
stat -c '%a %G' /srv/projects/acmu/shared/notes.md # 660 acmu
```

(d) The auditor cannot read the folder **[2 marks]**

Chandrima's access came from per-user entries on the files themselves, and those have been wiped. *Fix:* give him read-only access to everything under `/srv/projects/acmu/`, including files created there in future — *without* adding him to the group.

```
sudo -u chandrima cat /srv/projects/acmu/shared/notes.md # must work
sudo -u chandrima touch /srv/projects/acmu/shared/t2 # must FAIL
```

When all five are done

Buddhadev's original complaint needs (a), (b) and (c) together:

```
sudo -u buddhadev touch /srv/projects/acmu/shared/t3      # must work
sudo -u buddhadev cat /srv/projects/acmu/shared/notes.md  # must work
```

Constraints — these apply to every part, and are marked

- **Chandrima must not be put into the acmu group.** Audit policy forbids putting auditors into the groups they audit. He must be able to read and must *not* be able to write.
- No `chmod 777`, anywhere. This is a multi-user machine.
- Do not change anyone's login shell, password or home directory.

Hint. Group membership changes do not affect a shell that is already open; `sudo -u user` starts a fresh one.

Q3. Cannot log in to the ACMU lab server. [10 marks]

Reported by Debarshi Roy, research scholar: “I am supposed to log in to the lab server as `acmusrv` with the key the systems group set up for me, and run the nightly extract. It worked last week. Now it just says *Permission denied*, and then asks me for a password I was never given. I have not touched anything since.”

`ssh acmulab` should log you in to the account `acmusrv` on the lab server using the key `~/.ssh/id_acmu`, **without a password** — that account has no password to type.

Two things are wrong, and they produce the **same message**: `Permission denied (publickey)`. The first means the key is never offered at all; the second means it is offered and then refused. Fixing one and trying again will look like nothing happened, so work out which you are looking at before you change anything.

How to reproduce it

```
ssh acmulab

ssh -v acmulab 2>&1 | grep -i 'identity file\|offering\|publickey'
cat ~/.ssh/config
ls -l ~/.ssh/
```

What to fix [10 marks]

3.1 Make the acmulab alias use the right key. One line is missing from its block in `~/.ssh/config`; everything else in that block is correct.

3.2 Make the key usable. SSH refuses to use a private key that other people on the machine could read, and says so plainly under `-v`.

How to check you have fixed it

```
ssh -o BatchMode=yes acmulab whoami      # prints: acmusrv
ls -l ~/.ssh/id_acmu                     # -rw----- (0600)
```

`BatchMode=yes` forbids any password prompt, so if this prints `acmusrv` the key really is doing the work.

Hint. `ssh -v` is the whole diagnosis: it prints which key files it considered, which it offered, and what the server said. *Permission denied* means something answered and said no — quite different from *Connection refused*, which means nothing was listening at all.